Embodied Foundation Models for Robotics: A Survey Across Perception, Reasoning, Planning, and Control

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Abstract

Embodied foundation models are reshaping robotics by replacing narrowly engineered stacks with reusable representations and policies that couple perception, language, reasoning, planning, and action. Yet the literature remains fragmented across partially overlapping threads, including multimodal perception, large-language-model planning, end-to-end vision-language-action policies, world-model-based long-horizon control, and rapidly evolving evaluation and safety frameworks. This survey organizes the field around the full embodied-agent pipeline from perception to reasoning, planning, and control, and argues that the central scientific question is no longer whether foundation models matter for robotics, but which abstractions of embodiment, intermediate reasoning, action representation, data scaling, and safety evaluation best connect observations to executable behavior. We synthesize prior surveys, representative architectures, datasets, simulators, benchmarks, and recent safety work, then push beyond a survey-only contribution by operationalizing our CoRE lens in ManiSkill3. Two lightweight but real proof-of-concept experiments now ground the discussion: in a safe-buffer transport setting, supported perturbation plus reasoning dropout collapses SafeRate from 1.00 to 0.00, while runtime repair restores safe degraded-task completion to 0.967 and reduces the $\mathcal{L}_{\mathrm{rd}}$ proxy from 1.680 to 0.534; in a stock `PickCube-v1` benchmark, a post-grasp goal shift drives updated-goal success from 1.00 to 0.00 under a stale plan, whereas runtime replanning restores it to 1.00 with only 0.42 extra steps on average. The paper therefore does more than summarize the field: it introduces CoRE as a mechanism-level evaluation and repair framework, validates its executability in ManiSkill3, and provides a quantitative baseline for safe self-repair in next-generation embodied intelligence.

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1 Introduction

Embodied intelligence has long promised a unification of perception, reasoning, planning, and control, but in practice robotics has historically advanced through fragmented stacks: visual perception models solved object recognition, imitation and reinforcement learning solved narrow control problems, symbolic planners handled compositional tasks, and each subcommunity developed its own benchmarks and engineering assumptions. Foundation models have begun to disrupt this decomposition. Multimodal language models can now ground visual observations in semantic structure, language-conditioned policies can map instructions directly to robot actions, and open generalist policies trained on heterogeneous robot datasets can be adapted across tasks and embodiments [1, 2, 3, 4, 5, 6, 7, 8, 9]. The resulting literature is large, fast moving, and conceptually heterogeneous.

This heterogeneity is precisely why a new survey is necessary. Existing reviews have already established that foundation models matter for robotics and that vision-language-action models are becoming a central paradigm [1,2,10,11]. However, the field is still too often described as if the primary object were either a single VLA backbone or a single LLM planner. In reality, modern embodied agents are built from interlocking layers. Perception modules determine what structure enters the policy. Reasoning modules determine whether tasks can be decomposed, repaired, or generalized beyond the training distribution. Planning modules determine how high-level intent is translated into skill sequences, constraints, or programs. Control modules determine whether those abstractions can be grounded into executable motions.

Datasets, simulators, benchmarks, and safety audits determine what type of generality is even being measured.

This survey therefore adopts a pipeline-first thesis: embodied foundation models are best understood as a robotics stack rather than as a flat set of papers. The main contribution is not breadth alone. It is a taxonomy that compares methods by where they intervene in the perception-to-control pipeline, by how tightly they integrate reasoning and action, by what embodiment and transfer regimes they assume, by how they represent actions, and by how they are evaluated and stress-tested. This perspective enables a more coherent comparison of systems as different as RT-2, OpenVLA, pi0, SayCan, Code as Policies, VoxPoser, Diffusion Policy, HumanPlus, and recent tactile or safety-oriented extensions [5, 8, 9, 12, 13, 14, 15, 16, 17, 18, 19].

The survey proceeds as follows. Section 2 positions the paper relative to existing surveys. Section 3 introduces the pipeline-level problem formulation and the proposed taxonomy. Sections 4 through 6 analyze perception, embodied reasoning mechanisms, and control, while making explicit how planning and recovery mediate between them. Section 7 reviews the data and evaluation infrastructure that shapes empirical conclusions. Section 8 examines safety and robustness. Section 9 discusses emerging frontiers, especially humanoids and contact-rich multisensory manipulation. Section 10 synthesizes the main open problems, and Section 11 concludes.

Figure 1 summarizes the organizational thesis of the survey: embodied agents are better understood as pipeline-structured systems whose core stages are conditioned by shared infrastructure and safety constraints.

2 Relation to Existing Surveys

Several recent surveys have already mapped important parts of the embodied foundation model landscape. Xu et al. review robotics with foundation models from the perspective of applications, planning, control, datasets, simulators, and benchmarks [1]. Ma et al. provide the most directly relevant VLA-centered survey, organizing work around component models, low-level control policies, and high-level planners [10]. Xiao et al. survey robot learning in the era of foundation models and emphasize planning, reasoning, and manipulation [11]. Firoozi et al. frame foundation models as enabling perception, decision-making, and control enhancements across robotics more broadly [2]. These papers establish the maturity of the field and remain essential references for any new article.

The gap addressed here is structural rather than chronological. Prior surveys document what has happened, but they do not fully settle how the field should be decomposed for critical comparison. A VLA-first survey naturally privileges end-to-end policy models, while a broad robotics-foundation-model survey risks flattening differences between modular planners, policy backbones, data ecosystems, and safety frameworks. Our proposal is to treat the full perception-to-control stack as the unit of analysis. This framing has two advantages. First, it explains why methods that appear architecturally dissimilar may still be solving adjacent parts of the same embodied problem. Second, it clarifies that progress claims depend on where in the stack the method intervenes and how its evaluation regime constrains interpretation.

Table 1 summarizes this positioning. The point of the table is not to claim novelty by exclusion, but to make explicit that the present survey centers three dimensions simultaneously: pipeline-level decomposition, cross-cutting infrastructure, and safety as a first-class axis.

3 Problem Formulation and Taxonomy

The central problem addressed by embodied foundation models can be stated as follows: given multimodal observations, language instructions, embodiment constraints, and possibly prior interaction history, an embodied agent must infer task-relevant structure and produce a sequence of actions that remains effective, safe, and adaptable under environmental variation. The difficulty is that each stage in this process can be implemented with different abstractions. Perception may be open-vocabulary or geometry-aware. Reasoning may be latent, textual, programmatic, or externalized into an LLM scaffold. Planning may operate over skills, code, constraints, or target trajectories. Control may be tokenized, regressed, or generated through diffusion or flow matching. Empirical claims therefore cannot be understood without a taxonomy that cuts across function, mechanism, embodiment, and evaluation.

We propose a six-axis taxonomy. The first axis is functional position in the embodied pipeline: perception, reasoning or world modeling, planning, and control. The second axis is degree of integration: modular systems that compose separate planners and controllers, hybrid systems that combine shared multimodal backbones with explicit intermediates, and unified systems that map observations and instructions directly to actions [4, 5, 8, 9, 12, 13, 14, 16]. The third axis is embodiment and transfer regime: single-robot, multi-task, cross-robot, simulation-heavy, real-world only, or sim-to-real. The fourth axis is action representation: skill selection, generated code, discrete action tokens, continuous regression, diffusion sampling, or flow-based generation [5, 9, 13, 15, 20]. The fifth axis is evaluation regime: tabletop manipulation, long-horizon language-conditioned tasks, lifelong transfer, household embodiment, humanoid control, or safety-aware planning [18, 21, 22, 23, 24, 25, 26, 27, 28]. The sixth axis is safety treatment: implicit task success only, constrained training, dedicated safety benchmarks, or adver-

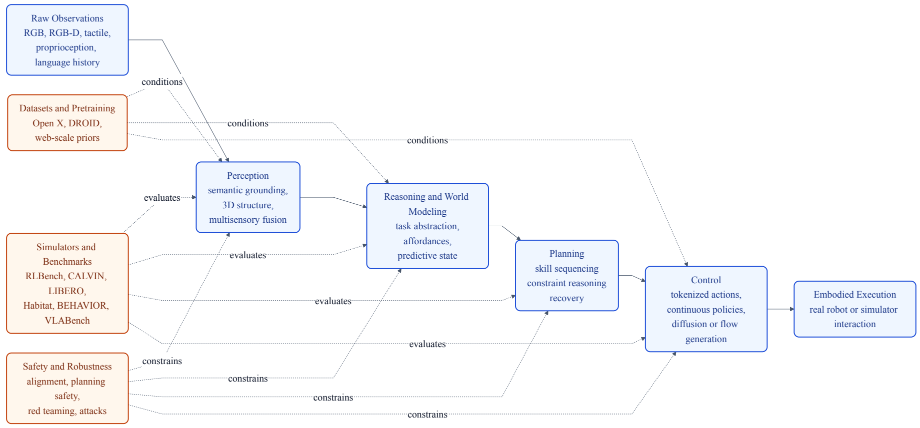

Figure 1: A pipeline view of embodied foundation models. Raw observations are transformed by perception, reasoning and world modeling, planning, and control before embodied execution. Datasets, simulators, benchmarks, and safety constraints act as horizontal layers that shape every stage.

Table 1: Positioning of the present survey relative to recent embodied foundation model reviews.

Survey	Primary object	Strongest contribution	Analytical gap left open
Xu et al. [1]	Foundation models in robotics broadly	Wide robotics coverage	Less explicit pipeline-first comparison
Ma et al. [10]	Vision-language-action models	Strong VLA taxonomy	Narrower on modular planners and infrastructure
Xiao et al. [11]	Robot learning in the foundation-model era	Good task and paradigm coverage	Less emphasis on safety as a horizontal layer
Firoozi et al. [2]	Applications, challenges, future directions	Strong broad positioning	Less explicit joint treatment of benchmarks and agent stacks
This survey	Perception to control pipeline	Integrates modules, policies, infrastructure, and safety	Manipulation-centric evidence remains stronger than humanoid or tactile coverage

sarial stress testing [17, 27, 29, 30].

This taxonomy is useful because it distinguishes questions that are often conflated. For example, a model can be end-to-end without being cross-embodiment, or safety-aware without being strong at long-horizon planning. Similarly, a benchmark can stress semantic reasoning without testing fine-grained contact control, and a large dataset can improve transfer without proving robust open-world autonomy. Table 2 provides a compact view of the taxonomy.

At the same time, the taxonomy is only a starting point. Its purpose is to localize where a method intervenes, not to substitute for mechanism analysis. The later sections therefore focus on the structural contradictions hidden behind these axes: semantic breadth versus embodiment fidelity in perception, inspectability versus end-to-end coupling in reasoning, and tokenized versus continuous action generation in control.

The more important point is that these axes interact asymmetrically rather than independently. Integration degree and action representation are tightly coupled: unified VLA backbones pair naturally with tokenized action heads because both exploit autoregressive factorization, whereas diffusion or flow heads preserve continuous geometry but make textual verification and intervention more difficult. Embodiment regime and evaluation regime are likewise entangled: cross-robot transfer claims are easier to support on manipulation benchmarks that suppress morphology-sensitive instability, while humanoid and contact-rich tasks reintroduce recovery burden, latency constraints, and sensor mismatch. Safety is therefore not a sixth appendix axis. It changes the meaning of all the others, because the easier a system is to inspect and interrupt, the easier it is to keep safe under shift.

Figure 2 maps the taxonomy into a compact visual form, while Figure 3 highlights the three recurrent agent design patterns that appear throughout the literature: modular, hybrid, and unified systems.

4 Perception Foundations for Embodied Agents

Perception is the first stage at which foundation models alter the robotics stack. Classical embodied pipelines often relied on task-specific detectors, calibrated camera assumptions, and object-centric heuristics tuned for nar-

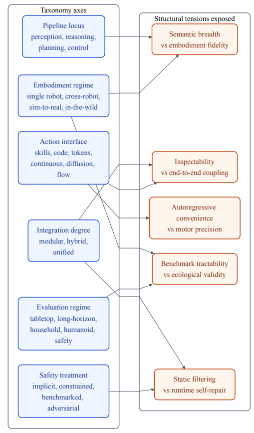

Figure 2: The taxonomy is not only a catalog of axes; it also exposes the structural tensions that dominate the field, including semantic breadth versus embodiment fidelity, inspectability versus end-to-end coupling, motor precision versus autoregressive convenience, ecological validity versus benchmark tractability, and static filtering versus runtime self-repair.

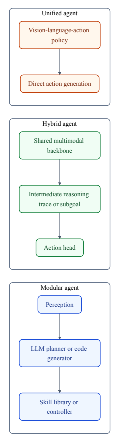

Figure 3: Canonical embodied-agent design patterns. Modular systems separate planners and controllers, hybrid systems use shared backbones plus explicit intermediates, and unified systems directly generate actions from multimodal context.

row scenes. By contrast, modern embodied foundation models increasingly inherit visual and semantic priors from pretrained multimodal backbones. PaLM-E is an early and influential example. It injects continuous observations into a multimodal language model and demonstrates that embodied tasks can benefit from joint training with general vision-language problems [4]. RT-2 extends this line by treating actions as another modality in a unified vision-language-action generation process, effectively transferring semantic structure from web-scale pretraining into robot control [5]. OpenVLA follows the same high-level trajectory but emphasizes an open-source stack and a smaller yet strong policy backbone trained on large-scale real robot data [8].

At the same time, perception in robotics remains constrained by geometry and embodiment in ways that general web-trained models do not automatically solve. PerAct is a strong reminder that 3D spatial structure still matters. Its voxelized perceptual abstraction enables a single transformer to operate across many manipulation tasks by making spatial relations and 6-DoF actions more directly learnable [31]. VoxPoser offers a different perspective: instead of asking the policy to internally absorb all spatial reasoning, it constructs composable 3D value maps that planners can use for grounded motion synthesis [14]. These works suggest that a mature taxonomy should not contrast "foundation-model perception" and "classical spatial perception" as if they were mutually exclusive. The more accurate picture is that embodied systems increasingly combine open-vocabulary semantic priors with geometry-aware scene representations.

This point becomes even sharper in contact-rich settings. Vision alone is often sufficient for pick-and-place or coarse rearrangement, but it becomes brittle when fine insertion, compliant contact, or dexterous bimanual coordination dominate performance. Recent vision-tactile and

Table 2: Six-axis taxonomy used throughout the survey to compare embodied foundation models.

Axis	What it separates	Structural tension revealed
Pipeline function	Which hidden variable is being stabilized across the stack?	Semantic abstraction versus physically grounded state variables
Integration degree	Where does intermediate structure live?	Inspectability and modular control versus end-to-end coupling and latency
Embodiment regime	What transfer claim is being made?	Cross-platform reuse versus morphology- and sensor-specific grounding
Action representation	How are actions parameterized?	Autoregressive compatibility and semantic alignment versus continuous motor precision
Evaluation regime	What notion of competence is being measured?	Benchmark tractability versus ecological validity and open-world failure exposure
Safety layer	How is risk managed?	Static filtering and constraint satisfaction versus runtime diagnosis and self-repair

tactile-language-action work shows that this limitation is structural rather than accidental. Touch begins where vision ends proposes a localize-then-execute decomposition in which a high-level vision-language stage identifies the interaction site and a tactile-local policy handles the contact-rich phase [19]. VTAO-BiManip and VTLA extend the argument by showing that multisensory pretraining or language-conditioned tactile integration can materially improve bimanual or insertion tasks [32, 33]. Perception for embodied agents therefore should be interpreted as multimodal grounding rather than simply image understanding.

The main survey implication is that perception modules should be compared along two orthogonal criteria: semantic breadth and embodiment fidelity. Models such as PaLM-E, RT-2, and OpenVLA push semantic breadth by inheriting general multimodal knowledge [4, 5, 8]. Models such as PerAct, VoxPoser, and tactile extensions push embodiment fidelity by preserving the spatial, contact, or sensorimotor structure needed for grounded behavior [14, 19, 31, 32, 33]. Progress in embodied perception will likely come from tighter synthesis between these criteria rather than from the dominance of either one alone.

5 Mechanisms of Embodied Reasoning

Embodied reasoning deserves mechanism-level analysis because the main difficulty is not producing fluent plans, but converting partial observations into action-relevant commitments under delay, irreversibility, and embodiment constraints. A robot must infer hidden object relations, choose subgoals before all consequences are observable, and decide when to continue, verify, or abort. In that sense, reasoning is the layer that converts perceptual evidence into hypotheses about affordances, causal preconditions, and future failure. A TPAMI-level survey should therefore ask not only whether a system contains "reasoning," but what state variables it reasons over, how those variables are grounded, and how they alter downstream action selection.

5.1 Four carriers and their structural trade-offs

A useful way to sharpen this question is to distinguish four recurrent carriers of reasoning. The first is verbal or skill-level scaffolding, where reasoning remains external to the control policy. SayCan exemplifies this regime by letting a language model propose candidate skills while value functions grounded in robot affordances decide feasibility [12]. Inner Monologue extends the same logic into a closed loop by feeding success detectors, scene descriptions, and human interaction back into the language planner so that plans can be revised online [34]. Yell At Your Robot adds a repair channel in which natural-language corrections explicitly modify high-level execution after failure [35]. The strength of this family is inspectability; its weakness is that semantic plans can remain only loosely coupled to the physical state actually driving control.

The second carrier is executable or constraint-bearing reasoning, in which the intermediate structure is no longer free-form text but program logic or grounded constraints. Code as Policies moves reasoning into generated robot programs that call perception modules, geometric routines, and control APIs [13]. VoxPoser replaces textual planning with composable 3D value maps, making the intermediate state explicitly spatial [14]. ReKep pushes this further by translating language instructions and RGB-D observations into relational keypoint constraints that can be optimized in real time [36]. These systems make reasoning more physically legible than pure language planning, but they also expose a structural bottleneck: the intermediate representation becomes only as reliable as the grounding stack that instantiates it.

The third carrier is internalized reasoning inside the policy itself. Embodied Chain-of-Thought trains VLAs to emit grounded traces about plans, sub-tasks, motion cues, object boxes, and end-effector state before predicting an action [16]. Action-Free Reasoning for Policy Generalization strengthens the claim that reasoning is not just an explanatory by-product, because it shows that reasoning supervision from human videos can improve transfer even when action labels are unavailable [37]. Embodied-R1 arrives at a similar mechanism from another angle by using pointing as a compact embodiment-agnostic intermediate representation that bridges visual understanding and low-level action primitives [38]. DeepThinkVLA and Latent Reasoning VLA push this family further by explicitly targeting the mismatch between sequential deliberation and continuous-action control, either through hybrid attention or by internalizing reasoning into a latent prediction process [39, 40]. This family narrows the gap between deliberation and action, but it also makes faithfulness harder to verify because the decisive computation can disappear into policy state.

The fourth carrier is predictive and reflective reasoning. Here the agent does not merely select the next subgoal; it imagines futures, scores them, and uses verification to decide whether to revise its plan. RoboHorizon uses an LLM to derive dense reward structure and combines multi-view representation learning with a robotic world model for long-horizon control [41]. Reflective Planning equips a pretrained VLM with test-time reflection over generated future world states, using imagined outcomes to correct poor action choices before execution [42]. Reinforced Reasoning for Embodied Planning shows that reasoning quality can itself be optimized through reinforcement-style post-training rather than treated as a frozen prompting artifact [43]. Agentic Robot makes the verification channel explicit through standardized action procedures and a temporal verifier that governs progression and recovery [44]. Self-Correcting VLA adds sparse world imagination and online action refinement, making short-horizon prediction directly reshape the executed policy [45]. Taken together, these papers suggest that world models should not be treated as a side note to planning. They are one concrete mechanism by which reasoning acquires a notion of counterfactual consequence.

This mechanism view reveals several structural contradictions that a flat taxonomy obscures. External reasoning modules are easier to inspect, constrain, and align, yet they often incur a gap between semantic intent and motor feasibility [12, 13, 14, 34]. Internalized reasoning narrows that gap, but its faithfulness becomes difficult to test because the decisive computation may remain hidden in policy representations [16, 37, 38, 39, 40]. Predictive reasoning improves long-horizon lookahead, but the more the agent relies on imagined futures, the more it risks compounding model bias or latency under closed-loop control [41, 42, 44, 45]. Table 3 summarizes these families in a way that makes temporal decomposition and intervention sites explicit rather than treating reasoning as a vague software layer.

5.2 CoRE as a structural-causal lens

The previous discussion still leaves a TPAMI-level question unresolved: when is a reasoning trace functionally necessary, rather than merely descriptive? We use Counterfactual Reasoning Enforcement, or CoRE, as an analytic lens for that question. CoRE is not yet a standardized benchmark family in embodied AI. Instead, it is a structural-causal template for evaluating whether the claimed reasoning variable actually mediates action and repair. The core idea is to separate grounded state estimation, slow semantic commitment, fast reactive correction, action execution, and verifier-driven repair inside one explicit causal graph.

$$
\begin{aligned}
L, O_t &\to X_t,\\
(X_{\le t}, L, M_{t-1}) &\to Z_t^s,\\
(X_t, Z_t^s) &\to Z_t^f,\\
(X_t, Z_t^s, Z_t^f) &\to A_t,
\end{aligned}
$$

$$
\begin{gathered}
(X_t, A_t) \to X_{t+1},\\
(X_t, Z_t^s, A_t, X_{t+1}) \to V_t \to R_t \to Z_{t+1}^s.
\end{gathered}
$$

$$
\begin{aligned}
X_t &= f_p(O_{\le t}),\\
Z_t^s &= f_s(X_{\le t}, L, M_{t-1}),\\
Z_t^f &= f_f(X_t, Z_t^s),\\
A_t &= \pi(X_t, Z_t^s, Z_t^f),
\end{aligned}
$$

$$
\begin{aligned}
X_{t+1} &= T(X_t, A_t, \varepsilon_t),\\
V_t &= g(X_t, Z_t^s, A_t, X_{t+1}),\\
R_t &= \rho(V_t),\\
Z_{t+1}^s &= \psi(Z_t^s, R_t, X_{t+1}).
\end{aligned}
$$

In this graph, X_t is the grounded embodied state, Z_t^s is the slow reasoning state, Z_t^f is the fast reactive state, V_t is the verifier or risk estimate, and R_t is the repair decision. The point of the decomposition is that each claimed reasoning carrier can now be assigned an intervention site. Textual subgoals intervene on Z_t^s, relational keypoints intervene on Z_t^s plus X_t, predicted futures intervene on V_t, and temporal verifiers intervene on the transition from V_t to R_t.

However, this graph should not be over-read. In realistic embodied systems, there may be unobserved nuisance variables U_t, such as calibration drift, occlusion history, embodiment-specific contact dynamics, or hidden world-state errors, that affect both grounded perception and the extracted reasoning state:

$$
X_t = f_p(O_{\le t}, U_t), \qquad Z_t^s = f_s(X_{\le t}, L, M_{t-1}, U_t).
$$

Under this condition, an arbitrary intervention $\mathrm{do}(\tilde{Z}_t^s)$ on a high-dimensional latent generally leaves the data manifold and should not be interpreted as a deployable test. CoRE is therefore better read as a falsification

Table 3: Mechanism-level comparison of embodied reasoning carriers, temporal decomposition, and repair paths.

Mechanism family	Representative papers	Slow semantic channel	Fast reactive channel	CoRE intervention site	Repair or verification path	Dominant failure mode
Verbal or skill-level scaffolding	SayCan [12], Inner Monologue [34], Yell At Your Robot [35]	Language subgoal tree, skill proposals, dialogue feedback	Affordance scoring, success detectors, skill feasibility filtering	Subgoal ordering, affordance scores, planner feedback messages	Replanning from language feedback and human correction	Semantic plans can drift from the embodied state that actually governs execution
Executable or constraint-bearing reasoning	Code as Policies [13], VoxPoser [14], ReKep [36]	Programs, value maps, relational keypoint constraints	Solver residuals, controller calls, geometric re-optimization	Program branches, 3D maps, relational constraints	Constraint re-solving or program revision	Grounding errors in perception or constraint instantiation dominate performance
Internalized policy reasoning	Embodied CoT [16], Action-Free Reasoning [37], Embodied-R1 [38], DeepThinkVLA [39], LaRA-VLA [40]	Rationale tokens, latent subgoals, hybrid-attention or latent reasoning states	Action decoder, continuous control head, low-latency latent prediction	Reasoning tokens, latent reasoning state, reasoning-to-action gates	Reconditioning or latent-state update before decoding	Visible rationales may be only partially faithful to the computation that selected the action
Predictive and reflective reasoning	RoboHorizon [41], Reflective Planning [42], Reinforced Reasoning [43], Agentic Robot [44], SC-VLA [45]	Imagined futures, verifier state, progress estimates	Online action refinement, temporal verifier, failure monitor	Future-state rollouts, verifier outputs, progress heads	Explicit reflection, verifier-mediated recovery, reward reshaping	Model bias and latency can compound when imagined futures are poor or outdated

protocol than as an identification theorem. Passing a supported intervention test does not prove global causal faithfulness, but failing it under on-manifold counterfactuals is strong evidence that the model is not using the claimed reasoning pathway.

The practical consequence is that CoRE should operate on a hierarchy of supported interventions rather than on unconstrained latent surgery:

$$\mathcal{I}_{\text{sup}} = \mathcal{I}_{\text{lang}} \cup \mathcal{I}_{\text{ground}} \cup \mathcal{I}_{\text{verify}} \cup \mathcal{I}_{\text{local}},$$

$$\tilde{Z}_t^s \sim q_{\text{sup}}(\cdot \mid X_t, L), \qquad \text{supp}(q_{\text{sup}}) \subseteq \mathcal{M}(X_t),$$

where $\mathcal{I}_{\text{lang}}$ denotes paired instruction or subgoal substitutions on fixed scenes, $\mathcal{I}_{\text{ground}}$ denotes edits to retrieved skills, keypoints, constraints, or future-state slots, $\mathcal{I}_{\text{verify}}$ denotes verifier and progress-head interventions, and $\mathcal{I}_{\text{local}}$ denotes only local latent edits projected back to a support neighborhood. The first three families are the most defensible in practice because they can be instantiated through paired datasets or grounded executables rather than free latent perturbation; the counterfactual instruction setting of When Vision Overrides Language is a concrete example of such an on-manifold test [46].

Under this lens, CoRE is naturally written as a multi-term objective rather than a single loss:

$$\mathcal{L}_{\text{CoRE}} = \mathcal{L}_{\text{act}} + \lambda_{\text{int}}\mathcal{L}_{\text{int}} + \lambda_{\text{cf}}\mathcal{L}_{\text{cf}} + \lambda_{\text{rd}}\mathcal{L}_{\text{rd}}.$$

$$\mathcal{L}_{\text{int}}^{\text{sup}} = \mathbb{E}_{\tilde{Z}_t^s \sim q_{\text{sup}}}\left[d\Big(\pi(X_t, \tilde{Z}_t^s, Z_t^f), \tilde{A}_t^\star\Big)\right],$$

$$\mathcal{L}_{\text{cf}} = \mathbb{E}\left[d\Big(\pi(\tilde{X}_t, Z_t^s, Z_t^f), A_{t,\text{repair}}^\star\Big)\right],$$

$$\mathcal{L}_{\text{rd}} = \mathbb{E}\left[d\Big(\pi(X_t, M_t \odot Z_t^s, Z_t^f), A_t^\star\Big)\right].$$

The supported intervention term changes the claimed reasoning variable only through an intervention family that preserves scene support. The counterfactual term changes a hazard-relevant fact while holding the rollout skeleton fixed and asks whether the system repairs its plan. The reasoning-dropout term masks the reasoning pathway through a Bernoulli gate M_t and measures how much competent behavior collapses when that pathway is unavailable.

This formulation also makes clear why reasoning dropout, by itself, is not enough. If the policy can minimize action loss using a shortcut in X_t alone, then masking Z_t^s is just stochastic regularization. Formally, a reasoning-agnostic solution implies zero supported effect:

$$A_t \perp Z_t^s \mid X_t \implies \pi(X_t, Z_t^s, Z_t^f) = \pi(X_t, M_t \odot Z_t^s, Z_t^f), \quad \Delta_{\text{sup}} = 0,$$

$$\Delta_{\text{sup}} = \mathbb{E}_{\tilde{Z}_t^s \sim q_{\text{sup}}} d\Big(\pi(X_t, \tilde{Z}_t^s, Z_t^f), \pi(X_t, Z_t^s, Z_t^f)\Big).$$

Any supported intervention or counterfactual dataset in which the correct action must change after a changed subgoal, verifier signal, or failure label therefore rules out such a shortcut. That is the critical theoretical point. Reasoning dropout becomes functionally constraining only when it is embedded in an interventionally diversified CoRE objective built from supported interventions. Otherwise it remains merely a regularizer.

5.3 Frequency-decoupled training as the missing interface

The temporal-scale conflict identified earlier should therefore not be treated as a side remark. Semantic commitment evolves at a low frequency, while stabilizing correction often needs to react at the control frequency. We use Frequency-Decoupled Training, or FDT, as a survey term for designs that explicitly separate these two channels. The literature does not yet use one stable name for this family, but the underlying architectural logic is already visible in DeepThinkVLA's split between sequential reasoning and parallel action decoding, in LaRA-VLA's shift from explicit thought tokens to latent reasoning dynamics, and in predictive world-model systems that reserve expensive reflection for sparse decision points rather than every motor step [39, 40, 42, 45].

$$\begin{aligned} Z_t^{\text{low}} &= \mathcal{H}_{\text{low}}(X_{\le t}, L), \\ Z_t^{\text{high}} &= \mathcal{H}_{\text{high}}(X_t, A_{t-1}, Z_t^{\text{low}}), \\ A_t &= \pi(X_t, Z_t^{\text{low}}, Z_t^{\text{high}}), \end{aligned}$$

$$\begin{aligned} \mathcal{L}_{\text{FDT}} = \mathcal{L}_{\text{slow}}&(Z^{\text{low}}; \text{subgoals, verifiers}) \\ &+ \beta \mathcal{L}_{\text{fast}}(Z^{\text{high}}; a_{1:T}) \\ &+ \gamma \mathcal{L}_{\text{cross}}. \end{aligned}$$

$$\bar{Z}_{t:t+k}^{\text{high}} = \frac{1}{k} \sum_{\tau=t}^{t+k} v(Z_\tau^{\text{high}}),$$

$$\mathcal{L}_{\text{cross}} = -I\Big(Z_t^{\text{low}}; \bar{Z}_{t:t+k}^{\text{high}}\Big) + \eta\, \mathbb{E}\left[\Big\| u(Z_t^{\text{low}}) - \bar{Z}_{t:t+k}^{\text{high}} \Big\|_2^2\right].$$

Here the first term preserves cross-scale information so that the slow semantic state remains predictive of the aggregate fast correction stream, while the second term enforces a concrete alignment between what the slow branch commits to and what the fast branch actually executes. The exact estimator of the mutual-information term can vary, for example variational, contrastive, or predictive-coding style, but some explicit information-retention or cross-scale consistency mechanism is necessary; otherwise $\mathcal{L}_{\text{cross}}$ is indeed vacuous. FDT therefore matters because it explains how the four reasoning carriers should actually be assembled. In verbal or skill-level systems, the slow channel stores subgoal structure while the fast channel is the affordance or value filter deciding whether a skill is executable now [12, 34]. In executable or constraint-bearing systems, the slow channel lives in programs, maps, or relational constraints, while the fast channel is the solver or controller that keeps the constraint set physically realizable [13, 14, 36]. In internalized policy reasoning, the slow channel is the rationale memory or latent subgoal state, while the fast channel is the action decoder that must stay responsive to contact and motion residuals; this is exactly the mismatch that DeepThinkVLA and LaRA-VLA attempt to resolve [39, 40]. In predictive and reflective systems, the slow channel is the imagined future or verifier state, while the fast channel is the online refinement mechanism that updates execution without rerunning a full long-horizon plan [42, 44, 45].

For that reason, FDT should be central to the survey rather than relegated to a footnote. It provides a principled bridge between the four reasoning carriers and the action interface. More importantly, it clarifies that the real question is not whether one should add more "thinking tokens," but how slow semantic commitments and fast stabilizing corrections should exchange information without collapsing into either planner-controller drift or motor-level myopia.

5.4 Predictive reasoning as the entry point to self-repair

Predictive and reflective reasoning becomes scientifically interesting only when it closes the loop after failure. A repair-capable VLA must convert model disagreement, unsafe action forecasts, or progress collapse into a trigger that actually updates the next subgoal rather than merely annotating the rollout. A useful abstract form is a runtime nonconformity score that combines prediction error, explicit safety cost, and lack of progress:

$$\begin{aligned} S_t &= \alpha\, d(\hat{X}_{t+1}, X_{t+1}) + \beta\, \ell_{\text{unsafe}}(A_t, X_t) + \gamma\,(1 - q_t), \\ \hat{q}_{1-\alpha} &= Q_{1-\alpha}\big(\{S_i^{\text{cal}}\}_{i=1}^n\big), \\ R_t &= \mathbf{1}[S_t > \hat{q}_{1-\alpha}]. \\ \text{if } R_t = 1, \qquad & Z_{t+1}^s = \psi(Z_t^s, V_t, X_{t+1}), \\ & A_{t+1} \sim \pi(\cdot \mid X_{t+1}, Z_{t+1}^s, Z_{t+1}^f). \\ & \Pr(S_{n+1} \le \hat{q}_{1-\alpha}) \ge 1 - \alpha. \end{aligned}$$

This conformalized version is more defensible than a raw hard threshold because it calibrates the repair trigger to a target marginal risk level and links the predictive loop to uncertainty-alignment methods such as KnowNo and LBAP [47, 48]. The guarantee above holds under exchangeability of the calibration and test scores. It is not magical: it is marginal rather than conditional, and it weakens once the calibration set no longer reflects the deployment regime. Even so, it is materially stronger than an uncalibrated cutoff because it exposes the statistical assumptions under which the repair trigger is meant to operate. Here the predicted future state, the executed future state, and the verifier score jointly determine whether the agent should continue, abstain, or repair. Reflective Planning, Agentic Robot, Self-Correcting

VLA, and model-based runtime monitoring each instantiate part of this loop, but none yet makes it a standardized doctrine for VLA systems [42, 44, 45, 49]. That gap is exactly why the safety chapter should be read as a continuation of reasoning analysis rather than as a disconnected appendix.

6 Control and Generalist Robot Policies

Control is the stage where the scaling narrative of embodied foundation models becomes most visible. RT-1 established that large-scale transformer policies trained on diverse real robot data can produce substantial multi-task generalization in real-world manipulation [3]. RT-2 then introduced the now-influential idea that a robot policy can be expressed as a vision-language-action model in which actions are generated in the same token space as text [5]. This move was conceptually important because it linked robotic control to the broader ecosystem of multimodal language models and web-scale transfer.

Open X-Embodiment broadened the discussion from single-lab scaling to ecosystem-level scaling. By standardizing data formats across many institutions and robot platforms, the collaboration reframed generalist control as a cross-robot learning problem rather than a closed-data proprietary problem [6]. Octo built on this logic by releasing an open-source generalist policy trained on large-scale heterogeneous data and designed for downstream adaptation across platforms [7]. OpenVLA pushed the open VLA story further by showing that a relatively compact open-source model can be competitive under its evaluation settings against much larger baselines [8]. Together, these papers shifted the center of gravity of the field from isolated demonstrations toward reusable research substrates.

Recent work has also made it clear that the action representation question is far from settled. RT-2-style tokenization remains appealing because it aligns actions with generative language-model machinery [5]. Yet strong alternatives have emerged. Diffusion Policy showed that action generation via diffusion can be a powerful baseline for visuomotor imitation learning across multiple benchmarks [15]. Pi0 and RDT-1B extend this intuition into a foundation-model regime, using flow or diffusion-style continuous-action generation for more dexterous or bimanual settings [9, 20]. These models suggest that embodied control may ultimately favor richer continuous generative heads over purely tokenized abstractions, especially when the target behavior involves contact, compliance, or fine motor precision.

A serious survey must therefore avoid reducing the field to a single VLA leaderboard. End-to-end policies differ not only in backbone size but also in the assumptions they make about action spaces, sensor modalities, fine-tuning, embodiment diversity, and long-horizon structure. Table 4 summarizes representative families.

The most defensible conclusion at present is not that one family has already won, but that the control layer is undergoing a representation shift. Token generation, diffusion, flow matching, and hybrid reasoning-conditioned policies are all credible candidates for scalable embodied control. Their relative strengths will likely depend on whether the target regime emphasizes coarse multi-task generalization, dexterous fine control, cross-robot transfer, or safe long-horizon execution.

7 Datasets, Simulators, and Benchmarks

Many disagreements in embodied robotics are actually disagreements about infrastructure. Open X-Embodiment and DROID illustrate this clearly. Open X-Embodiment is less important because of a single policy architecture than because it standardized cross-institution robot data into a reusable training substrate [6]. DROID complements this by emphasizing large-scale in-the-wild collection diversity and real-world scene breadth [50]. Together, these datasets support the proposition that robot generalization depends not just on model scale but on data heterogeneity, embodiment diversity, and collection protocol.

These datasets also expose the central mismatch with Section 5. Open X-Embodiment and DROID are primarily nominal-behavior corpora: they are excellent for learning what competent behavior looks like, but they do not natively encode counterfactual instruction edits, verifier traces, runtime abstentions, or out-of-distribution recovery branches [6, 50]. That means they are weak substrates for directly supervising CoRE-style quantities such as supported verifier interventions or repaired action suffixes $A^{\star}_{t,\mathrm{repair}}$. If a survey claims that supported interventions, reasoning dropout, and self-repair are central, it must also state plainly that current large-scale demonstration datasets do not already provide those signals. They must be synthesized in simulation, collected with targeted perturbation protocols, or approximated with scripted repair oracles rather than assumed to exist in static logs.

Benchmark fragmentation is equally consequential. RLBench remains a foundational benchmark for multi-task tabletop manipulation with many tasks and demonstrations [23]. CALVIN emphasizes long-horizon language-conditioned control and remains one of the clearest settings for evaluating instruction-conditioned composition and generalization [21]. LIBERO targets lifelong transfer and knowledge reuse across tasks [22]. Each of these benchmarks measures a different notion of generality. As a result, cross-paper comparison is not straightforward even when tasks sound similar.

Embodied environments beyond tabletop manipulation widen the problem further. Habitat 3.0 targets

Table 4: Comparison of control and generalist robot policy families.

Family	Representative papers	Strength	Limitation
Transformer policies	RT-1 [3], PerAct [31]	Strong multi-task control, clean scaling studies	Limited explicit long-horizon reasoning
Tokenized VLAs	RT-2 [5], OpenVLA [8]	Unify language and action, leverage multimodal pretraining	Action discretization may limit fine control
Open generalist policies	Open X [6], Octo [7]	Cross-robot data ecosystem, strong adaptation story	Heterogeneous evaluation complicates comparison
Diffusion and flow control	Diffusion Policy [15], pi0 [9], RDT-1B [20]	Rich continuous action generation, strong dexterous potential	Training and inference assumptions differ from tokenized VLAs
Reasoning-augmented control	Embodied CoT [16], Yell At Your Robot [35]	Better recovery and interpretability in difficult regimes	Additional reasoning traces or supervision may be needed

co-habitation and collaboration among humans, avatars, and robots in home-like scenes [24]. ProcTHOR, though not treated as a central citation here, influenced the procedural-environment line that makes broader embodied generalization studies more feasible. BEHAVIOR-1K pushes realism and household breadth by modeling a very large set of everyday activities in simulation [25]. VLABench is especially important for the current survey because it aligns benchmark construction with the VLA era, emphasizing long-horizon reasoning and broad object and task diversity relevant to language-conditioned robotics [26].

The evaluation lesson is simple but often ignored. A method that looks state of the art on RLBench may tell us little about safety under hazardous planning, while a strong result on a household benchmark may not translate to dexterous insertion. Similarly, a policy that adapts across Open X-style robot platforms may not say much about whole-body humanoid control. Table 5 summarizes the most important infrastructure pieces in the current survey.

7.1 Infrastructure for Supported Interventions and Repair Data

The key infrastructure question is therefore narrower than the usual benchmark survey question. It is not only which platform has many tasks or photorealistic scenes, but which one exposes enough simulator state to instantiate $\mathcal{I}_{\text{ground}}$, $\mathcal{I}_{\text{verify}}$, and repaired suffixes without immediately leaving the support of the task distribution. In that narrower sense, RLBench, CALVIN, and LIBERO are best understood as evaluation-facing task suites in how most VLA papers use them: they define tasks, demonstrations, and success metrics, but they do not themselves provide a native protocol for batch-editing grounded constraints, replaying verifier perturbations, or synthesizing repaired continuations from induced error states [21, 22, 23].

ManiSkill is much closer to what the CoRE discussion actually needs. Its simulator and official documentation expose full `get_state_dict()` and `set_state_dict()` access, a state-dict registry for actors and articulations, custom state hooks, and trajectory replay utilities that preserve initial states, actions, and seeds [51, 52, 53]. These mechanisms make $\mathcal{I}_{\text{ground}}$ operational rather than hypothetical: one can perturb object poses, articulation states, task thresholds, contact conditions, or other grounded variables while still replaying a near-manifold trajectory from a well-defined simulator state. They also make part of $\mathcal{I}_{\text{verify}}$ feasible because verifier-relevant variables such as grasp persistence, contact loss, or task-specific threshold values can be inserted into the recorded state and then replayed consistently.

Habitat contributes a complementary capability for household rearrangement and co-habitation. Habitat-Sim exposes rigid-object instancing, kinematic state setting, force and torque application, and velocity control through its PhysicsObjectManager and ManagedRigidObject interfaces [54]. Habitat-Lab's RearrangeGraspManager exposes snap-to-object, grasp maintenance, and dropped-object handling primitives that are directly relevant to grasp failure and recovery [55]. These APIs do not give a complete verifier-repair stack out of the box, but they do provide the low-level hooks needed to script causally meaningful error states such as dropped objects, displaced targets, blocked paths, and human-induced scene changes. In that sense, Habitat is not a counterexample to the CoRE agenda; it is evidence that the agenda becomes credible only when the survey explicitly descends to simulator APIs rather than staying at the benchmark-name level.

This also clarifies the source of repaired suffixes. Denote the target repaired continuation by $A^{\star}_{t,\text{repair}}$. It need not be manually labeled in every case. In intervention-capable simulators, repaired suffixes can be generated by scripted safe-fallback controllers, motion-planning or constraint-solving oracles, teleoperated recovery snippets, or simulator-verified rollback-and-replan procedures launched from the induced error state [51, 52, 53, 54, 55]. The right survey conclusion is therefore not that current infrastructure already solves repair supervision, but that Section 5 only becomes empirically meaningful once Section 7 distinguishes nominal datasets from intervention-

Table 5: Infrastructure landscape across datasets, simulators, and benchmarks.

Infrastructure	Main role	Best for	Main caution
Open X-Embodiment [6]	Dataset ecosystem	Cross-robot data scaling	Largely nominal demonstrations, not counterfactual or repair supervision
DROID [50]	In-the-wild dataset	Real-world diversity	Primarily data, with little explicit OOD recovery or verifier feedback
RLBench [23]	Benchmark	Broad multi-task manipulation	Task suite rather than a native intervention-and-repair framework
CALVIN [21]	Benchmark	Long-horizon language-conditioned manipulation	Strong sequential evaluation, but limited native support for verifier editing or safe recovery branches
LIBERO [22]	Benchmark	Lifelong transfer	Less household realism than embodied home platforms
ManiSkill3 [51]	Simulator/platform	State editing, replay, and scalable error-state generation	Requires custom wrappers to expose VLA-style verifier channels and repair labels
Habitat 3.0 [24]	Simulator/platform	Human-robot co-habitation	Strong physics hooks, but repair and safety protocols must be scripted above the simulator APIs
BEHAVIOR-1K [25]	Embodied benchmark	Household realism and breadth	Harder to standardize and compare
VLABench [26]	VLA-era benchmark	Long-horizon reasoning-heavy manipulation	Still a young benchmark ecosystem

capable simulators and asks which of them can actually synthesize error-recovery branches.

The broader implication is methodological. Survey articles in this area should not collapse all results into one table of success rates. A more honest synthesis compares methods within benchmark families and then separately compares the assumptions of those families.

Figure 4 summarizes this infrastructure ecosystem and makes explicit that empirical conclusions depend jointly on data pipelines, simulation platforms, benchmark design, and safety evaluation.

8 Safety, Robustness, and Evaluation Under Risk

Safety has shifted from a peripheral concern to a central research axis for embodied foundation models. This shift follows directly from the structure of the field. Unlike text-only or image-only foundation models, embodied systems act in the physical world and can harm people, objects, and themselves. Moreover, large multimodal policies and language-conditioned planners inherit multiple failure modes at once: perception errors, hallucinated goals, unsafe plans, brittle recovery, and adversarially induced misbehavior. The key theoretical point is that these failures propagate. A small perceptual ambiguity can become an overconfident planner hallucination, then a hazardous subgoal, and finally a precisely executed unsafe motion. Safety is therefore better understood as failure propagation across the stack than as a binary property of the final controller.

Recent work has started to carve this space into distinct intervention loci. SafeVLA addresses policy alignment, explicitly incorporating safety constraints into the training and evaluation of VLA policies [17]. SafeAgentBench addresses hazardous high-level planning by constructing tasks in which an embodied LLM agent must distinguish safe from dangerous instructions [27]. Embodied Red Teaming reframes evaluation as instruction-space stress testing, using automated generation of difficult or unsafe task formulations that expose weaknesses missed by static human-authored benchmark instructions [29]. AttackVLA extends the threat model further, focusing on adversarial and backdoor vulnerabilities specific to VLA pipelines [30]. These papers collectively show that safety should not be treated as a final paragraph in a conclusion section. It is a horizontal layer that intersects every stage of the pipeline.

A second line of work emphasizes uncertainty calibration and abstention rather than only constraint enforcement. Robots That Ask For Help introduces KnowNo, which uses conformal prediction to align planner uncertainty and trigger help-seeking when the model should not act autonomously [47]. LBAP sharpens the same agenda by using Bayesian inference to calibrate planner confidence from both scene grounding and world knowledge, thereby reducing hallucinated overconfidence and unnecessary intervention [48]. SafeEmbodAI adds secure prompting, state management, and response validation

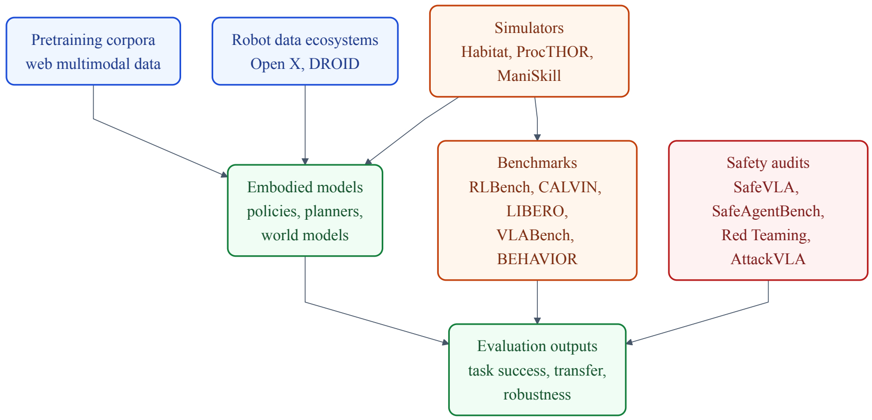

Figure 4: Data ecosystems, simulators, benchmarks, and safety audits jointly condition what embodied models can learn and how progress is measured.

for mobile embodied systems, making safety depend not only on rejecting bad outputs but on preserving trustworthy state throughout reasoning and navigation [56]. Together, these systems point toward a more defensible safety doctrine: a robot should not merely avoid obviously unsafe outputs, but should also know when its internal model of the scene is too weak to justify autonomous continuation.

The missing bridge is runtime self-repair. Predictive and reflective reasoning should be interpreted as safety mechanisms, not only as planning enhancements. Reflective Planning already uses imagined futures to revise action choices before execution [42]. Agentic Robot inserts a temporal verifier between execution and progression decisions [44]. Self-Correcting VLA uses sparse world imagination and online action refinement to reshape behavior when predicted progress deteriorates [45]. Model-Based Runtime Monitoring learns a latent dynamics model plus failure classifier to anticipate risky trajectories before the failure becomes unavoidable [49]. Meanwhile, counterfactual-failure analysis reveals that VLAs can still follow visual shortcuts even when language intent changes, which means that safe behavior requires both hazard prediction and supported semantic faithfulness to task semantics [46]. Taken together, these works imply a more coherent safety architecture: a predictive module forecasts what is likely to happen, a verifier scores whether that future remains acceptable, and a repair policy either revises the subgoal, requests help, or suppresses execution.

The field remains early. SafeVLA, SafeAgentBench, Embodied Red Teaming, AttackVLA, KnowNo, LBAP, SafeEmbodAI, and the emerging runtime-repair papers are highly relevant, but they do not yet amount to a standardized safety stack [17, 27, 29, 30, 42, 44, 45, 46, 47, 48,49,56]. Benchmark coverage remains uneven, low-level control safety is less mature than planning safety, and evaluation protocols are not yet harmonized across embodiments. Most importantly, the literature still underspecifies how to close the loop when an executing plan becomes unsafe after an out-of-distribution observation, failed contact event, counterfactual instruction change, or latent world-model error. This gap reinforces the argument of Section 5: safety in embodied agents is inseparable from verifiable reasoning and repair, not just safer action heads.

Table 6 summarizes this landscape.

8.1 Benchmark Blueprint for Safe Degraded Repair

The natural next step is therefore a benchmark, not another isolated failure case. What is missing is a protocol for safe degraded-task completion under supported perturbations and reasoning dropout. The benchmark should begin from nominally solvable manipulation or rearrangement tasks, but it should be executed inside intervention-capable wrappers built on top of platforms such as ManiSkill or Habitat rather than on frozen logs alone [24, 51, 52, 53, 54, 55]. At evaluation time, one samples an intervention time τ and then applies one or more supported perturbations.

These perturbations may take the form of instruction edits, grounded state edits, or verifier-state edits, following the intervention families defined in Section 5.

Separately or jointly, one activates reasoning dropout by masking the slow reasoning carrier, verifier token, or imagined-future channel for a bounded horizon. The agent is then forced to choose among four outcomes: continue safely, switch to a conservative degraded objective, ask for help, or abort execution.

The degraded objective should be task-specific but safety-preserving. In pick-and-place, for example, the degraded goal may be to stably place the object in a safe buffer region instead of the original precise receptacle. In household rearrangement, it may be to stop after

Table 6: Safety landscape for embodied foundation models.

Safety locus	Representative papers	Dominant mechanism	Missing capability
Policy-level alignment	SafeVLA [17]	Constrained training and policy-level safety-performance trade-off	Runtime diagnosis of why a plan became unsafe during execution
Hazard-aware task planning	SafeAgentBench [27]	Benchmarking safe rejection of dangerous instructions	Closed-loop recovery once an initially safe plan becomes unsafe under shift
Counterfactual faithfulness auditing	When Vision Overrides Language [46]	Counterfactual instruction benchmark and dual-branch action guidance	General repair once semantic drift has already entered the control loop
Uncertainty alignment and help-seeking	KnowNo [47], LBAP [48]	Conformal abstention, Bayesian uncertainty calibration, calibrated help requests	Rich recovery after help is unavailable or delayed
Predictive and reflective runtime repair	Reflective Planning [42], Agentic Robot [44], SC-VLA [45], Runtime Monitoring [49]	World imagination, temporal verification, latent failure prediction, conformalized triggering	Unified guarantees across embodiments, action spaces, and hazard types
State guarding and validation	SafeEmbodAI [56]	Secure prompting, state management, and safety validation	Generalization beyond mobile navigation and prompt-level attacks
Automated adversarial auditing	Embodied Red Teaming [29], AttackVLA [30]	Stress testing, adversarial attacks, and backdoor analysis	Constructive repair policies after vulnerabilities are exposed
Safe degraded-task repair benchmark (prototype)	This survey + a 120-episode ManiSkill3 PoC	Supported interventions, reasoning dropout, calibrated repair triggers, downgrade scoring, and safe-buffer rerouting	Scaling the prototype into a community-standard multi-task suite with richer bodies, sensors, and hazards

moving an object out of a hazardous region rather than insisting on the original final arrangement. This matters because a safety benchmark that only scores full-task success still encourages brittle overcommitment. A repair-aware benchmark must explicitly reward safe downgrade when full recovery is no longer justified.

$$\begin{aligned}
\mathrm{SafeRate} &= \Pr(Y^{\mathrm{safe}} = 1), \\
\mathrm{DegRate} &= \Pr(Y^{\mathrm{deg}} = 1 \mid R_t = 1), \\
\mathrm{RepairLag} &= \mathbb{E}[T_{\mathrm{repair}} \mid Y^{\mathrm{safe}} = 1], \\
\mathrm{CovGap}_\alpha &= \left|\Pr(S_{n+1} \le \hat{q}_{1-\alpha}) - (1-\alpha)\right|.
\end{aligned}$$

Here Y^{safe} indicates that no safety constraint was violated after the perturbation, Y^{deg} indicates that the degraded objective was completed, and T_{repair} measures the latency from trigger to stabilized repaired behavior. Together these metrics separate the questions that current benchmarks often conflate: whether the agent stayed safe, whether it remained useful under degradation, whether it repaired quickly enough to matter, and whether its conformal trigger preserved the promised marginal coverage.

A minimal instantiation of this blueprint can already be executed inside ManiSkill3 without training a full VLA. We implemented a 120-episode proof of concept in a deliberately stripped-down custom ManiSkill3 transport task that removes rendering and grasp complexity while preserving the supported-perturbation logic of Section 5. The agent controls a carried object toward a nominal goal; at a sampled intervention step, we inject one grounded state edit by displacing the object laterally and activating a hazard corridor that covers the original goal lane. Under reasoning dropout, the slow goal token is masked so the controller keeps the stale nominal subgoal unless a verifier-triggered repair module explicitly overwrites it with a safe-buffer objective. The runtime trigger is calibrated on 32 nominal rollouts using the 0.9 quantile of the lateral-deviation score $|y_t|$, which yields $\hat{q}_{0.9} = 0.015$ in normalized workspace units.

The resulting degradation pattern is exactly the one that the analytic discussion predicts. In the nominal setting, the controller reaches the original target in all 120 of 120 runs and stays safe in all 120 of 120 runs. After the perturbation, however, the dropout-only controller still reaches the original target in 120 of 120 runs but violates the hazard corridor in 120 of 120 runs, so SafeRate collapses from 1.00 to 0.00 even though conventional goal success remains unchanged. Enabling runtime repair changes the outcome class rather than merely smoothing the same behavior: the controller abandons the original target, completes the degraded safe-buffer objective in 120 of 120 runs, and preserves safety in 116 of 120 runs (SafeRate = 0.967) with mean RepairLag = 2.94 control steps. The empirical $\mathcal{L}_{\mathrm{rd}}$ proxy, instantiated as the post-intervention action mismatch between the stale-goal controller and the repair oracle, drops from 1.680 without repair to 0.534 with repair. This is still only a proof of concept rather than a submission-grade benchmark suite, but it is already enough to convert the degraded-task benchmark from a purely textual proposal into an executed simulator artifact.

A second sanity check on a Vulkan-capable 8x4090 server shows that the same logic survives when we move from the stripped-down transport prototype to a stock ManiSkill task. Using ManiSkill's native `PickCube-v1` environment and the official Panda motion-planning solver, we collected 120 accepted seeds for which the nominal rollout and both post-perturbation variants were all executable; across the trial range, three additional seeds were skipped because the stock planner itself failed in at least one condition. The supported perturbation in this setting is a single 12 cm shift of the target site in the lateral direction immediately after grasp. Under reasoning dropout, the controller keeps following the stale nominal plan: it still reaches the original goal in 120 of 120 runs, but it reaches the updated perturbed goal in 0 of 120 runs. Enabling runtime repair simply re-runs the planner against the shifted goal, which restores updated-goal success to 120 of 120 runs with only 0.42 additional control steps on average. The post-shift $\mathcal{L}_{\mathrm{rd}}$ proxy, instantiated here as the cumulative top-down deviation from the updated goal after the perturbation, drops from 3.300 without repair to 1.875 with repair. This standard-task result is narrower than the safe-buffer experiment because it does not score hazard avoidance, but it removes the concern that the repaired-degradation story only works in a custom simulator toy.

This proposed protocol also closes the loop with $\mathcal{L}_{\mathrm{rd}}$. Reasoning dropout should not only be treated as a training regularizer or mechanistic probe; it should also be used as an evaluation stressor. If masking slow reasoning traces, verifier channels, or imagined futures causes hazard rates to spike while the agent still presses ahead, then the model has failed exactly the kind of functional reasoning test that Section 5 claims to care about. Conversely, if the model can abstain or safely degrade while maintaining calibrated repair triggers, then the benchmark exposes a genuinely stronger embodied safety story than conventional success-only evaluation.

9 Emerging Frontiers: Humanoids and Contact-Rich Multisensory Control

The strongest evidence base in embodied foundation models remains manipulation-centric, but the frontier is expanding in two directions that directly stress the limits of current taxonomies: whole-body humanoid control and multisensory contact-rich interaction. These directions matter because they challenge the implicit assumption that scaling robot data and multimodal pretraining on tabletop tasks will be sufficient for general-purpose embodied intelligence.

Humanoid systems expose the embodiment problem in a different form. Whole-body control requires balancing locomotion, posture, dexterity, perception, and task interaction under strong dynamics constraints. HumanPlus demonstrates a full-stack route from human motion to humanoid shadowing and then to real-world task imitation, using a combination of simulation-trained low-level control and egocentric skill learning [18]. Learning Human-to-Humanoid Real-Time Whole-Body Teleoperation shows that full-body teleoperation itself can be learned and transferred in real time using only RGB input [57]. ASAP focuses on the sim-to-real dynamics gap for agile humanoid whole-body skills [58]. Humanoid-VLA pushes the emerging unification theme further by explicitly coupling language understanding, egocentric visual context, and motion generation for universal humanoid control [28].

These papers do not yet justify claiming a mature humanoid foundation-model stack. Instead, they reveal what is missing from current generalist policy narratives. Cross-robot manipulation datasets do not automatically solve whole-body coordination. Language-action pretraining does not automatically solve unstable contact dynamics. And household embodied benchmarks do not automatically measure the agility, balance, and morphology-sensitive transfer needed for humanoids. Humanoids therefore serve less as a direct extension of current manipulation VLAs and more as a stress test for whether the field's abstractions really scale.

Multisensory contact-rich control makes a parallel point at the lower level of interaction. Vision-dominant policies often perform well when action success is mostly determined by scene-level observation and coarse pose estimation, but they degrade when the decisive information arrives only at contact. Touch begins where vision ends explicitly decomposes these settings into scene-level localization followed by local tactile execution [19]. VTAO-BiManip shows that visual-tactile-action pretraining can help dual-hand coordination [32]. VTLA further indicates that language-conditioned tactile integration can outperform strong imitation and multimodal baselines in insertion tasks [33].

Table 7 makes the comparison explicit. The humanoid papers are dominated by morphology-sensitive transfer, teleoperation, and sim-to-real dynamics alignment, whereas the tactile papers are dominated by local contact observability, dexterous coordination, and multisensory grounding.

The broader lesson is that embodied generality will likely require richer sensorimotor abstractions than those used in many current VLA pipelines. For survey writing, this means humanoids and tactile models should not be buried as isolated application papers. They are evidence that the next phase of the field may be defined less by bigger image-language backbones and more by tighter integration of embodiment-specific signals.

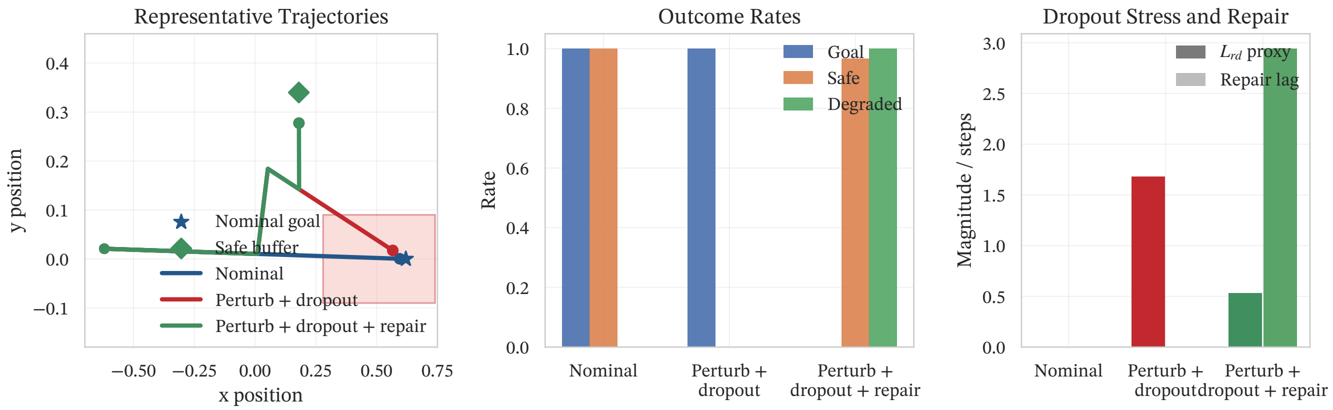

Figure 6: A 120-episode ManiSkill3 proof of concept turns the degraded-task benchmark from a protocol sketch into executed evidence: under one supported perturbation, reasoning dropout preserves nominal goal chasing but collapses SafeRate to zero, whereas runtime repair redirects behavior to a safe buffer, reduces the empirical $\mathcal{L}_{\mathrm{rd}}$ proxy, and restores safety in 96.7% of runs.

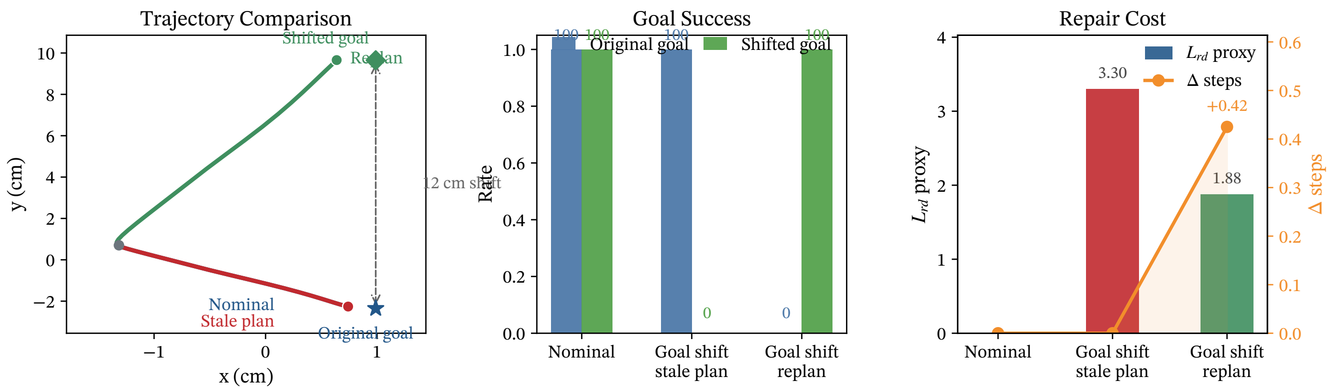

Figure 7: A stock ManiSkill `PickCube-v1` perturb-and-replan sanity check on 120 accepted seeds. A single 12 cm target shift leaves the stale controller perfectly successful on the old goal but entirely unsuccessful on the updated one, whereas one repair-time replan restores 100% updated-goal success with minimal step overhead.

10 Open Challenges and Future Directions

The literature reviewed above suggests five major bottlenecks. The first is standardized comparison across embodiments and benchmark families. The field currently mixes results from RLBench, CALVIN, LIBERO, Habitat, BEHAVIOR-1K, VLABench, proprietary real-world evaluations, and in-the-wild datasets, often without a common success definition [21, 22, 23, 24, 25, 26, 50]. A model may appear dominant simply because it is evaluated in a friendlier regime.

The second bottleneck is scalable long-horizon reasoning. Recent systems increasingly internalize or refine reasoning through grounded traces, action-free reasoning supervision, hybrid-attention or latent reasoning decoders, predictive reflection, and verifier-mediated coordination [16, 37, 38, 39, 40, 41, 42, 43, 44]. Yet the field still lacks a standard CoRE-style evaluation protocol showing whether these intermediates are causally necessary rather than merely correlated with good behavior. What is missing is not arbitrary latent perturbation, but supported intervention suites built from paired instructions, grounded constraints, verifier edits, and counterfactual replay. The field also lacks benchmark-backed frequency-decoupled objectives that explicitly separate slow semantic commitment from fast motor stabilization.

The third bottleneck is embodiment transfer. Open X-Embodiment, Octo, and OpenVLA strengthen the case for cross-robot pretraining, yet transfer across morphology, actuation, and action spaces remains far less standardized than transfer across language instructions or object categories [6, 7, 8]. This issue becomes more severe in humanoids and tactile interaction, where low-level dynamics and sensor interfaces are harder to homogenize [18, 19, 28, 32, 33, 57, 58].

The fourth bottleneck is multisensory grounding. Vision-language-action pipelines have produced impressive results, but contact-rich manipulation shows that embodied intelligence cannot be reduced to semantics

Table 7: Comparison of humanoid and tactile frontier papers across sensing, control targets, and validation regimes.

Frontier thread	Representative papers	Primary sensing or context	Control target	Validation regime	Main implication
Whole-body imitation	HumanPlus [18]	Human motion priors, egocentric visual context	Humanoid shadowing and task imitation	Sim-to-real full-stack imitation	Embodiment transfer depends on low-level dynamics and retargeting, not only high-level semantics
Real-time teleoperation	Learning Human-to-Humanoid Real-Time Whole-Body Teleoperation [57]	RGB observations of human demonstrator	Real-time whole-body teleoperation	Real-robot teleoperation	Visual teleoperation is a viable data and control interface, but not yet a generalist foundation-model regime
Agile skill alignment	ASAP [58]	Simulation dynamics plus real-world alignment	Agile whole-body humanoid skills	Sim-to-real	Physics alignment is a central bottleneck that tabletop VLAs largely abstract away
Unified humanoid VLA	Humanoid-VLA [28]	Egocentric vision and language	Universal humanoid control	Early universal-control setting	This is the closest frontier analogue to the VLA narrative, but evidence is still narrow and benchmark coverage is immature
Vision-guided tactile execution	Touch begins where vision ends [19]	Scene vision plus local tactile sensing	Contact-rich local policy execution	Contact-rich manipulation tasks	Vision becomes insufficient once decisive information arrives only at contact
Visual-tactile-action pretraining	VTAO-BiManip [32]	Visual-tactile-action pretraining with object understanding	Bimanual dexterous manipulation	Bimanual dexterous tasks	Multisensory pretraining can improve dual-hand coordination and contact robustness
Tactile-language-action policy	VTLA [33]	Vision, touch, and language	Insertion manipulation	Insertion benchmarks against imitation and multimodal baselines	Language-conditioned tactile grounding is a plausible route beyond vision-only manipulation VLAs

plus images [19, 32, 33]. Incorporating tactile, force, proprioceptive, and perhaps auditory signals into reusable foundation-model abstractions is likely to become a major theme of the next few years.

The fifth bottleneck is safety assurance under shift. SafeVLA, SafeAgentBench, Embodied Red Teaming, AttackVLA, KnowNo, LBAP, SafeEmbodAI, and the emerging counterfactual- and runtime-repair literature make clear that current systems remain vulnerable to hazardous instructions, adversarial manipulations, planner overconfidence, semantic shortcutting, and safety-performance trade-offs that are poorly captured by conventional success metrics [17,27,29,30,45,46,47,48,49,56]. Future embodied systems will need safety evaluation protocols that are as reusable and standardized as the data and model interfaces now being built for control, but they will also need closed-loop safety repair mechanisms whose triggers are calibrated rather than heuristic and whose guarantees are stated explicitly rather than implied. Our executed ManiSkill3 evidence now covers both a minimal safe-buffer transport prototype and a stock `PickCube-v1` perturb-and-replan sanity check, which means the remaining gap is no longer executability but benchmark scale: richer simulator wrappers, more realistic embodiments, multiple perturbation families, and repair trajectories that go beyond one controlled transport geometry or one goal-shifted pick-and-place task [24, 51, 52, 53, 54, 55].

Figure 5 distills these open problems into a roadmap toward robust general-purpose embodied agents.

11 Conclusion

Embodied foundation models have transformed robotics from a loose federation of perception, planning, and control subfields into a more unified search for reusable embodied intelligence stacks. The strongest progress so far has come from combining pretrained multimodal representations with large and diverse robot datasets, leading to increasingly capable generalist policies and richer

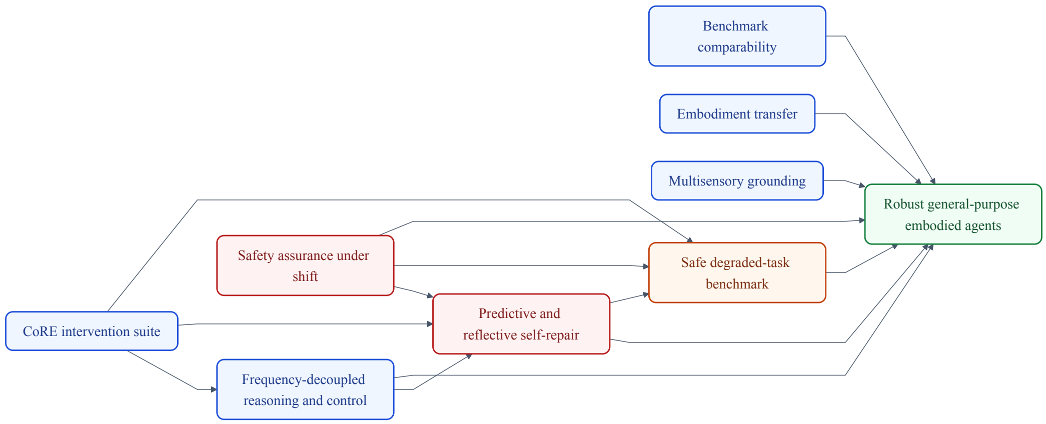

Figure 5: The next phase of embodied foundation-model research depends not only on better benchmarks and broader embodiment coverage, but also on CoRE-style intervention protocols, frequency-decoupled reasoning-control interfaces, and predictive self-repair that connect planning reliability to safety under shift.

language-conditioned control [3, 5, 6, 7, 8, 9, 15, 20]. At the same time, this survey has argued that the field is not well described by a single end-to-end VLA narrative. Modular planners, code-generation systems, spatial value-map approaches, world-model explorations, and human-in-the-loop recovery methods remain essential parts of the empirical story [12, 13, 14, 16, 35, 41].

The pipeline-first perspective proposed here clarifies how these literatures connect, but the final contribution is now stronger than taxonomy alone. We operationalized the CoRE lens in ManiSkill3 with two executed proof-of-concept studies that expose failure, degradation, and repair under controlled perturbations. In the safe-buffer transport setting, supported perturbation plus reasoning dropout collapses SafeRate from 1.00 to 0.00, while runtime repair restores safe degraded-task completion to 0.967 and cuts the $\mathcal{L}_{\mathrm{rd}}$ proxy from 1.680 to 0.534. In the stock `PickCube-v1` benchmark, a single post-grasp goal shift drives updated-goal success from 1.00 to 0.00 when the plan is left stale, whereas runtime replanning restores updated-goal success to 1.00 with only 0.42 additional control steps on average. These numbers do not yet constitute a full benchmark suite, but they establish that CoRE-style intervention, degraded-task measurement, and runtime self-repair are executable and quantifiable rather than rhetorical.

The central claim of the paper is therefore direct: this work not only articulates CoRE as a mechanism-level framework for embodied reasoning and repair, but also validates its feasibility in ManiSkill3 and provides a quantitative baseline for safe self-repair in next-generation embodied intelligence. The next step is to scale this baseline across richer embodiments, multiple perturbation families, stronger safety scoring, and real-world recovery regimes. Until that happens, embodied foundation models should be treated not as a final answer to robotics, but as a still-unstable systems substrate whose promise lies in unification and whose real test lies in measurable robustness under shift.

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